Parameter-free optimization

Лекции по оптимизации

Постановка

Задача:

$$\min_{x \in \mathbb{R}^d} f(x),$$

где f — **негладкая**., f - выпуклая.

Доступная информация: субградиенты

$$g(x) \in \partial f(x).$$

Типичное предположение для негладкого анализа:

$$|f(x) - f(y)| \leq M\|x - y\| \ \forall x, y \in \mathbb{R}^d,$$

то есть f M-Липшицева.

Из этого следует

$$\|g(x)\| \leq M \ \forall g(x) \in \partial f(x).$$

Постановка

Baseline: Subgradient Descent

$$x^{t+1} = x^t - \gamma^t g^t.$$

При правильно выбранных шагах

$$\gamma^t \equiv \gamma = \frac{\|x^0 - x^*\|}{M\sqrt{T}}$$

даёт сходимость

$$f(\bar{x}^T) - f(x^*) = \mathcal{O}\left(\frac{M\|x^0 - x^*\|}{\sqrt{T}}\right), \quad \bar{x}^T = \frac{1}{T}\sum_{t=1}^{T} x^t.$$

От adaptive методов к parameter-free идее

Классический AdaGrad-Norm stepsize:

$$\gamma^t = \frac{1}{\sqrt{\sum_{i=0}^{t} \|g^i\|^2}}$$

- Автоматически подстраивается под ландшафт функции потерь $f(x)$ (Липшицевость).
- Позволяет получать независимость шага от константы M.

$$\gamma^t = \frac{\|x^0 - x^*\|}{\sqrt{\sum_{i=0}^{t} \|g^i\|^2}}$$

Основная цель parameter-free оптимизации — адаптивность к $\|x^0 - x^*\|$.

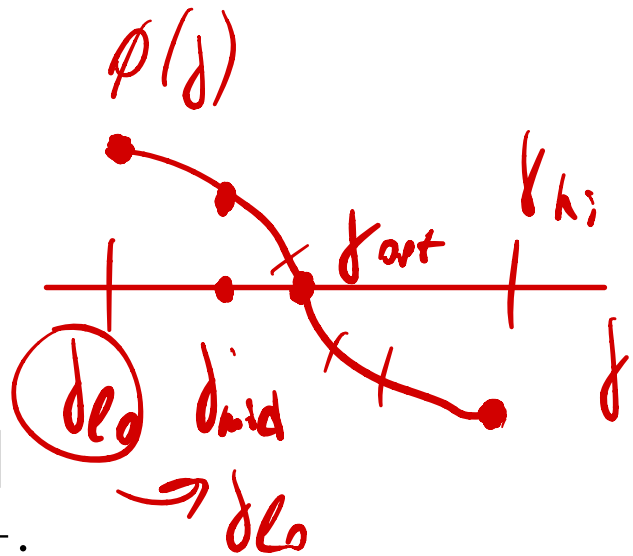
Интуиция

- **Наивный подход:** предварительный поиск шага ближе к оптимальному $\rightarrow$ запуск оптимайзера с найденным шагом.
- **Продвинутый подход:** поитерационная аппроксимация шага к начальному приближению $D = \|x^0 - x^*\| \rightarrow$ использование этой аппроксимации в шаге γ^t.

Наивный подход: Bisection

Задаем желаемый константный шаг для алгоритма γ_{opt}:

$$\gamma_{\text{opt}} = \frac{\max\limits_{0 \le i \le T-1} \|x^0 - x^i\|}{\sqrt{\sum\limits_{i=0}^{T-1} \|g^i\|^2}}.$$



Но! γ_{opt} зависит от γ_{opt}.
Положим

$$\phi(\gamma) = \frac{\max\limits_{0 \le i \le T-1} \|x^0 - x^i(\gamma)\|}{\sqrt{\sum\limits_{i=0}^{T-1} \|g^i(\gamma)\|^2}}.$$

Хотим найти: $\gamma \approx \phi(\gamma)$.

Наивный подход: Bisection [5]

Алгоритм 1 BISECTION procedure

Input: Optimal stepsize value $\phi(\gamma)$, lower stepsize bound $\gamma_{\mathbf{lo}}$, upper stepsize bound $\gamma_{\mathbf{hi}}$, $x^{\mathbf{0}} \in \mathbb{R}^d$, number of iterations T
$\phi(\gamma)$ (it is always in the form $\phi(\gamma) = \frac{\mathfrak{N}_T(\gamma)}{\mathfrak{D}_T(\gamma)}$)
if $\gamma_{\mathbf{hi}} \leqslant \phi(\gamma_{\mathbf{hi}})$ **then return** ∞ ▷ early infinite termination
end if
if $\gamma_{\mathbf{lo}} > \phi(\gamma_{\mathbf{lo}})$ **then return** $\gamma^*_{\mathbf{lo}} = \gamma_{\mathbf{lo}}$ ▷ early non-infinite termination
end if
while $\gamma_{\mathbf{hi}} > 2\gamma_{\mathbf{lo}}$ **do**
    $\gamma_{\mathbf{mid}} = \sqrt{\gamma_{\mathbf{lo}}\gamma_{\mathbf{hi}}}$
    $\mathfrak{N}_T(\gamma_{\mathbf{mid}}), \mathfrak{D}_T(\gamma_{\mathbf{mid}}) \leftarrow$ OPTIMIZER$(x^{\mathbf{0}}, T, \gamma_{\mathbf{mid}})$
    if $\gamma_{\mathbf{mid}} \leqslant \phi(\gamma_{\mathbf{mid}})$ **then**
        $\gamma_{\mathbf{lo}} = \gamma_{\mathbf{mid}}$
    else
        $\gamma_{\mathbf{hi}} = \gamma_{\mathbf{mid}}$
    end if ▷ bisection invariants: $\gamma_{\mathbf{lo}} < \phi(\gamma_{\mathbf{lo}})$, $\gamma_{\mathbf{hi}} > \phi(\gamma_{\mathbf{hi}})$
end while ▷ bisection stop condition: $\gamma_{\mathbf{hi}} \leqslant 2\gamma_{\mathbf{lo}}$
if $\mathfrak{N}_T(\gamma_{\mathbf{hi}}) \leqslant \mathfrak{N}_T(\gamma_{\mathbf{lo}}) \frac{\phi(\gamma_{\mathbf{hi}})}{\gamma_{\mathbf{hi}}}$ **then return** $\gamma^*_{\mathbf{hi}} = \gamma_{\mathbf{hi}}$ ▷ $\gamma_{\mathbf{hi}}$ return condition
elsereturn $\gamma^*_{\mathbf{lo}} = \gamma_{\mathbf{lo}}$ ▷ $\gamma_{\mathbf{lo}}$ return condition
end if

Наивный подход: Bisection

Стоимость процедуры Bisection:

$$\mathcal{O}\left(\log\log\frac{\gamma_{\text{hi}}}{\gamma_{\text{lo}}}\right).$$

Итоговая сходимость:

$$\tilde{\mathcal{O}}\left(\frac{DM}{\sqrt{T}}\right).$$

$\delta_{\text{opt.}} \to \underline{\mathcal{O}\left(\frac{DM}{\sqrt{T}}\right)}$

- \+ parameter-free
- \+ правильная оценка
- \- **долго**
- \- **непрактично**

Продвинутый подход: DoG [11]

Цель: дизайн алгоритма без использования дополнительных поисков/рестартов

Идея: давайте модифицируем шаг из предыдущего подхода так, чтобы использовать только известные на текущий момент аппроксимации:

$$\gamma^t = \frac{\max\limits_{0 \leq i \leq t} \|x^0 - x^i\|}{\sqrt{\alpha \sum\limits_{i=0}^{t} \|g^i\|^2}}.$$

Продвинутый подход: DoG

cat?!?

Что за alpha? Почему она возникла? Где здесь parameter-free?

$$d^t = \max_{i \le t} \|x^i - x^*\|$$

$$\bar{r}^t = \max_{i \le t} \|x^i - x^0\| \overset{?}{\le} D$$

Для простоты:

$$\gamma^t = \frac{\bar{r}^t}{\alpha \sqrt{\sum_{i=0}^{T-1} \|g^i\|^2}}$$

Convexity:

$$\langle g^i, x^i - x^* \rangle \overset{(1)}{\ge} f(x^i) - f(x^*) \ge 0$$

Step:

$$\|x^{i+1} - x^*\|^2 = \|x^{i+1} - x^i + x^i - x^*\|^2 =$$

$$= \|x^i - x^*\|^2 - 2\gamma^i \underbrace{\langle g^i, x^i - x^* \rangle}_{\ge 0 \ (1)} + (\gamma^i)^2 \|g^i\|^2$$

$$\sum_{i=0}^{j-1} \hookrightarrow \quad \|x^j - x^*\|^2 = \|x^0 - x^*\|^2 + \sum_{i=0}^{j-1} (\gamma^i)^2 \|g^i\|^2 =$$

$$\le \|x^0 - x^*\|^2 + \frac{(\bar{r}^j)^2 \sum_{i=0}^{T-1} \|g^i\|^2}{\alpha^2 \sum_{i=0}^{T-1} \|g^i\|^2} \quad \text{Сл (МТ)}$$

$$d^t$$

$$\max_{j \le t} \|x^j - x^*\|^2 = \|x^0 - x^*\|^2 + \frac{\max_{j \le t} \left(\max_{i \le j} \|x^0 - x^i\| \right)^2}{\alpha^2} =$$

$$= \|x^0 - x^*\|^2 + \frac{\left(\max_{i \le t} \|x^0 - x^i\| \right)^2}{\alpha^2} = \|x^0 - x^*\|^2 + \frac{(\bar{r}^t)^2}{\alpha^2} \quad (2)$$

$$\bar{r}^t = \max_{i \le t} \|x^0 - x^i\| = \|x^0 - x^k\| \le \|x^0 - x^*\| + \|x^* - x^k\| \le \quad (k \le t)$$

$$\le \|x^0 - x^*\| + \max_{j \le t} \|x^* - x^j\| \quad (3)$$

$$(2) \le \underbrace{\|x^0 - x^*\|^2}_{\|D} + \frac{1}{\alpha^2} \left(\underbrace{\|x^0 - x^*\|}_{\|D} + \underbrace{\max_{i \le t} \|x^j - x^*\|}_{\|d^t} \right)^2$$

$$\left(d^+ - \frac{1}{\alpha-1} D \right)^2 \le \frac{\alpha^2}{(\alpha-1)^2} D^2$$

$$\Downarrow$$

$$d^+ \le \frac{\alpha+1}{\alpha-1} D$$

$$(3) \Rightarrow \bar{r}^+ \le D + d^+ \le \frac{2\alpha}{\alpha-1} D$$

$$\alpha \neq 1$$

$$O\left(\frac{MD}{\sqrt{T}} \cdot \ln(\dots) \cdot \ln(\dots) \right)$$

Итого:

- Сходимость: $\widetilde{\mathcal{O}}\left(\frac{DM}{\sqrt{T}}\right)$, но под $\widetilde{\mathcal{O}}\,()$ скрыта α, которая теперь — дополнительный log-фактор.
- В Bisection сходимость также зависит от α, но α — константа больше 1.

Продвинутый метод: D-Adapation [8]

Идея:

- Модифицировать аппроксимацию D таким образом, чтобы она была ограничена сверху самим D.
- Использовать для аппроксимации непосредстсвенно шаги алгоритма (субградиенты).

Продвинутый метод: D-Adapation

Алгоритм 2 D-Adaptation

Input: $d^0, x^0, s^0 = 0$
for $t = 0, \ldots, T-1$ **do**
  $\lambda^t = \frac{d^t}{\sqrt{\sum_{i=0}^{t} \|g^i\|^2}}$
  $s^{t+1} = s^t + \lambda^t g^t$
  $\hat{d}^{t+1} = \frac{\|s^{t+1}\|^2 - \sum_{i=0}^{t} (\lambda^i)^2 \|g^i\|^2}{2\|s^{t+1}\|}$
  $d^{t+1} = \max(\hat{d}^{t+1}, d^t)$
  $x^{t+1} = x^t - \lambda^t g^t$
end for

$$s^{t+1} = s^t + \lambda^t g^t, \quad d^{t+1} = \frac{\|s^{t+1}\|^2 - \sum_{i=0}^{t} (\lambda^i)^2 \|g^i\|^2}{2\|s^{t+1}\|}$$

Лемма 1: (как оценить d^T?)

$$\frac{1}{2}\|s^{t+1}\|^2 = \frac{1}{2}\|s^t\|^2 - \lambda^t \langle s^t, g^t \rangle + \frac{1}{2}(\lambda^t)^2 \|g^t\|^2$$

$$-\sum_{t=0}^{T-1} \lambda^t \langle s^t, g^t \rangle = \frac{1}{2}\sum_{t=0}^{T-1} \left[\|s^t\|^2 - \|s^{t+1}\|^2\right] + \frac{1}{2}\sum_{t=0}^{T-1} (\lambda^t)^2 \|g^t\|^2$$

$$= -\frac{1}{2}\|s^T\|^2 + \frac{1}{2}\underbrace{\|s^0\|^2}_{0} + \frac{1}{2}\sum_{t=0}^{T-1} (\lambda^t)^2 \|g^t\|^2 \qquad (1)$$

Convexity:

$$\sum_{t=0}^{T-1} \lambda^t \left(f(x^t) - f(x^*)\right) \le \sum_{t=0}^{T-1} \lambda^t \langle g^t, x^t - x^* \rangle$$

$$= \sum_{t=0}^{T-1} \lambda^t \langle g^t, \underbrace{x^t - x^0}_{\text{// step}} \rangle + \sum_{t=0}^{T-1} \lambda^t \langle g^t, x^0 - x^* \rangle \quad = \quad (=)$$

$$-\sum_{i=0}^{t-1} \lambda^i g^i \overset{\text{s-update}}{=} s^t - s^0 = s^t$$

$$= \sum_{i=0}^{t-1} \left(s^i - s^{i+1}\right) =$$

$$= -s^t$$

$$(=) \quad -\sum_{t=0}^{T-1} \lambda^t \langle g^t, s^t \rangle + \langle s^T, x^0 - x^* \rangle$$

$$\le \|s^T\| D - \sum_{t=0}^{T-1} \lambda^t \langle g^t, s^t \rangle$$

Подст 1) →

$$\sum_{t=0}^{T-1} \lambda^t \left(f(x^t) - f(x^*) \right) \le \|S^T\| D + \frac{1}{2} \sum_{t=0}^{T-1} (\lambda^t)^2 \|g^t\|^2 - \frac{1}{2} \|S^T\|^2$$

$f(x^t) - f(x^*) \ge 0$

$$\Rightarrow \quad 0 \|S^T\|^2 \ge \frac{1}{2} \|S^T\|^2 - \frac{1}{2} \sum_{t=0}^{T-1} (\lambda^t)^2 \|g^t\|^2$$

$$D \ge \hat{d}^T = \frac{\|S^T\|^2 - \sum_{t=0}^{T-1} (\lambda^t)^2 \|g^t\|^2}{2 \|S^T\|}$$

□

Лемма 2: (Как оценить $\|S^T\|^2$?)

$$\text{By def.:} \quad \frac{1}{2} \|S^T\|^2 - \frac{1}{2} \sum_{t=0}^{T-1} (\lambda^t)^2 \|g^t\|^2 =$$

$$= \hat{d}^T \|S^T\| \overset{\text{мах}}{\le} d^T \|S^T\|$$

Let $\alpha^2 = 2 (d^T)^2$, $\beta^2 = \frac{1}{2} \|S^T\|^2$

$$2 \alpha \beta = 2 d^T \|S^T\| \le \alpha^2 + \beta^2 = 2 (d^T)^2 + \frac{1}{2} \|S^T\|^2$$

$$\le 2 (d^T)^2 + d^T \|S^T\| + \frac{1}{2} \sum_{t=0}^{T-1} (\lambda^t)^2 \|g^t\|^2$$

$$\|S^T\|^2 \le 2 d^T + \frac{1}{2 d^T} \sum_{t=0}^{T-1} (\lambda^t)^2 \|g^t\|^2$$

□

Th. / (convergence)

$$\sum_{t=0}^{T-1} \lambda^t \left(f(x^t) - f(x^*) \right) \overset{L1}{\leq} D \|g^T\| + \frac{1}{2} \sum_{t=0}^{T-1} (\lambda^t)^2 \|g^t\|^2$$

$$\overset{L2}{\leq} 2 D d^T + \frac{D}{2 d^T} \sum_{t=0}^{T-1} (\lambda^t)^2 \|g^t\|^2 + \frac{1}{2} \sum_{t=0}^{T-1} (\lambda^t)^2 \|g^t\|^2$$

$$= 2 D d^T + \frac{1}{2} \left(\frac{D}{d^T} + 1 \right) \sum_{t=0}^{T-1} (\lambda^t)^2 \|g^t\|^2 \leq$$

$$\lambda^t = \frac{d^t}{\sqrt{\sum_{i=0}^{t} \|g^i\|^2}}$$

$$\leq (d^T)^2$$

$$\leq 2 D d^T + \frac{D}{d^T} \sum_{t=0}^{T-1} \frac{(d^t)^2 \|g^t\|^2}{\sum_{i=0}^{t} \|g^i\|^2} \leq$$

$$\leq 2 D d^T + D d^T \sum_{t=0}^{T-1} \frac{\|g^t\|^2}{\sum_{i=0}^{t} \|g^i\|^2} \leq 6 D d^T \log(MT)$$

$$\leq 4 \log \left(\sum_{t=0}^{T-1} \|g^t\|^2 \right) \leq 4 \log(MT).$$

$$\frac{\sum_{t=0}^{T-1} \lambda^t \left(f(x^t) - f(x^*) \right)}{\sum_{t=0}^{T-1} \lambda^t} \leq \frac{6 D d^T \log(MT)}{\sum_{t=0}^{T-1} \frac{d^t}{\sqrt{\sum_{i=0}^{t} \|g^i\|^2}}} \leq M \sqrt{T}$$

$$\leq \frac{6MD\, d^T \sqrt{T}\, \log(MT)}{\sum_{t=0}^{T-1} d^t} \leq$$

Non-AS., $d^t \leq d^{t+1}$

$$\geq \frac{d^T T}{\log(d_T/d_0)}$$

$$\rightarrow \leq O\left(\frac{MD \log(MT) \log\left(\frac{d_T}{d_0}\right)}{\sqrt{T}}\right)$$

AS. $d^t \leq D \rightarrow$

$$\rightarrow \exists \hat{t}: d^t \geq \frac{1}{2} d^T \quad \forall t \geq \hat{t}, T \geq 2\hat{t}$$

$$\Rightarrow \sum_{t=0}^{T-1} d^t \geq \sum_{t=\hat{t}}^{T} d^t \geq$$

$$\geq \frac{1}{2} \sum_{t=\hat{t}}^{T} d^T = \frac{1}{2}(T-\hat{t}+1) d^T$$

$$\geq \frac{1}{4}(T+1) d^T$$

$$\rightarrow \leq O\left(\frac{MD \log(MT)}{\sqrt{T}}\right)$$

□

Продвинутый метод: D-Adaptation

Итого:

- Мы сделали ограниченную аппроксимацию.
- Нашли кейс, в котором избавились от дополнительного log-фактора.
- Можем ли мы улучшить сходимость с точки зрения log-фактора?

Продвинутый метод: Prodigy [15]

D-Adaptation stepsize:

$$\lambda^t = \frac{d^t}{\sqrt{\sum_{i=0}^{t} \|g^i\|^2}}.$$

Prodigy stepsize:

$$\lambda^t = \frac{(d^t)^2}{\sqrt{\sum_{i=0}^{t} (d^t)^2 \|g^i\|^2}}.$$

Продвинутый метод: Prodigy [15]

D-Adaptation Non-Asymptotic case:

$$\frac{d^T}{\sum_{t=0}^{T-1} d^t} \leq 4\frac{\log\left(\frac{d^T}{d^0}\right)}{T}.$$

Prodigy Non-Asymptotic case:

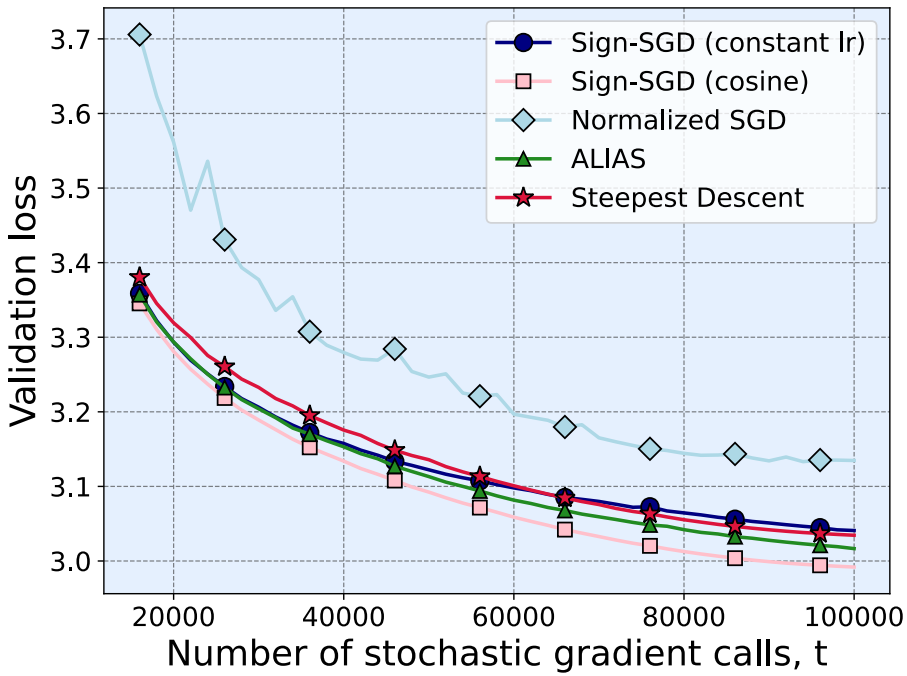
$$\frac{d^T}{\sqrt{\sum_{t=0}^{T-1} (d^t)^2}} \leq 4\frac{\sqrt{\log\left(\frac{d^T}{d^0}\right)}}{T}.$$

Итого: убрали корень из log-фактора.

Практические приложения

Почему Bisection плохо?

- Метод дает константный шаг.



130 M

Figure: Comparison on LLaMa pre-training.

Практические приложения

Ограниченность шага D-Adaptation.

Шаг D-Adaptation:

$$\gamma^t = \frac{d^t}{\sqrt{\sum_{i=0}^{t} \|\nabla f(x^i)\|_2^2}},$$

$\bar{r}^t = \max_{i \le t} \|x^0 - x^i\|$

где d^t построена так, что $d^t \leqslant \|x^0 - x^*\|_2$ и $\lim\limits_{t\to\infty} d^t \geqslant \frac{\|x^0 - x^*\|_2}{1+\sqrt{3}}$.

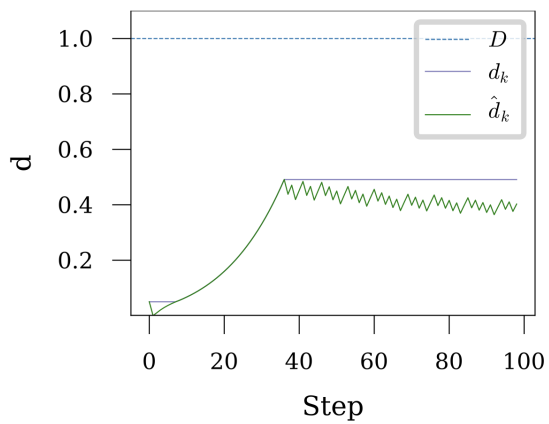


Figure: D-Adaptation stepsize.

Практические приложения

На практике используют Adam-версии алгоритмов.

Algorithm 5 Adam with D-Adaptation

Input: x_0,
d_0 (default 10^{-6}),
γ_k (default 1),
$\beta_1, \beta_2, \epsilon$ (default 0.9, 0.999, 10^{-8}).
$s_0 = 0, m_0 = 0, v_0 = 0, r_0 = 0$
for $k = 0$ **to** n **do**
$g_k \in \partial f(x_k, \xi_k)$
$m_{k+1} = \beta_1 m_k + (1 - \beta_1) d_k \gamma_k g_k$
$v_{k+1} = \beta_2 v_k + (1 - \beta_2) g_k^2$
$A_{k+1} = \operatorname{diag}(\sqrt{v_{k+1}} + \epsilon)$
$x_{k+1} = x_k - A_{k+1}^{-1} m_{k+1}$
Learning rate update
$s_{k+1} = \sqrt{\beta_2} s_k + (1 - \sqrt{\beta_2}) d_k \gamma_k g_k$
$r_{k+1} = \sqrt{\beta_2} r_k + (1 - \sqrt{\beta_2}) d_k \gamma_k \langle g_k, s_k \rangle_{A_{k+1}^{-1}}$
$\hat{d}_{k+1} = \dfrac{r_{k+1}}{(1 - \sqrt{\beta_2}) \|s_{k+1}\|_1}$
$d_{k+1} = \max(d_k, \hat{d}_{k+1})$
end for

Algorithm 3 Prodigy (Adam version)

Input: $d_0 > 0$ (default 10^{-6}), x_0, β_1 (default 0.9), β_2 (default 0.999), ϵ (default 10^{-8}), γ_k (default 1 with cosine annealing)
$r_0 = 0, s_0 = 0, m_0 = 0, v_0 = 0$
for $k = 0$ **to** n **do**
  $g_k \in \partial f(x_k)$
  $m_{k+1} = \beta_1 m_k + (1 - \beta_1) d_k g_k$
  $v_{k+1} = \beta_2 v_k + (1 - \beta_2) d_k^2 g_k^2$
  $r_{k+1} = \sqrt{\beta_2} r_k + (1 - \sqrt{\beta_2}) \gamma_k d_k^2 \langle g_k, x_0 - x_k \rangle$
  $s_{k+1} = \sqrt{\beta_2} s_k + (1 - \sqrt{\beta_2}) \gamma_k d_k^2 g_k$
  $\hat{d}_{k+1} = \dfrac{r_{k+1}}{\|s_{k+1}\|_1}$
  $d_{k+1} = \max(d_k, \hat{d}_{k+1})$
  $x_{k+1} = x_k - \gamma_k d_k m_{k+1} / (\sqrt{v_{k+1}} + d_k \epsilon)$
end for

Практические приложения

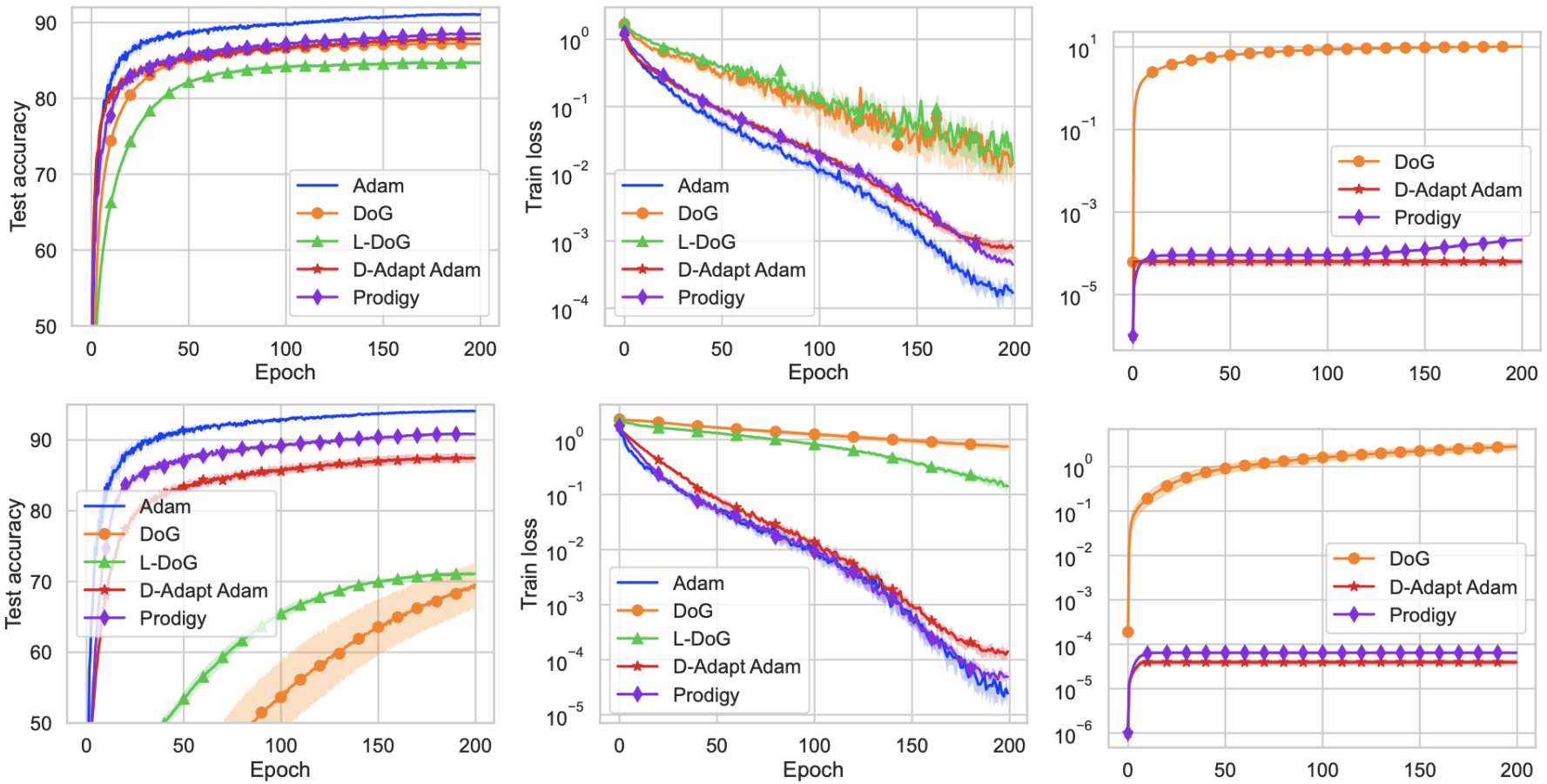


Figure 2: VGG11 and ResNet-50 training on CIFAR10. Left: test accuracy (%), middle: train loss, right: step sizes. “Prodigy” is used as given in Algorithm 3. As expected, Prodigy estimates a larger step size than D-Adaptation, which helps it reach test accuracy closer to the one of Adam.

Практические приложения

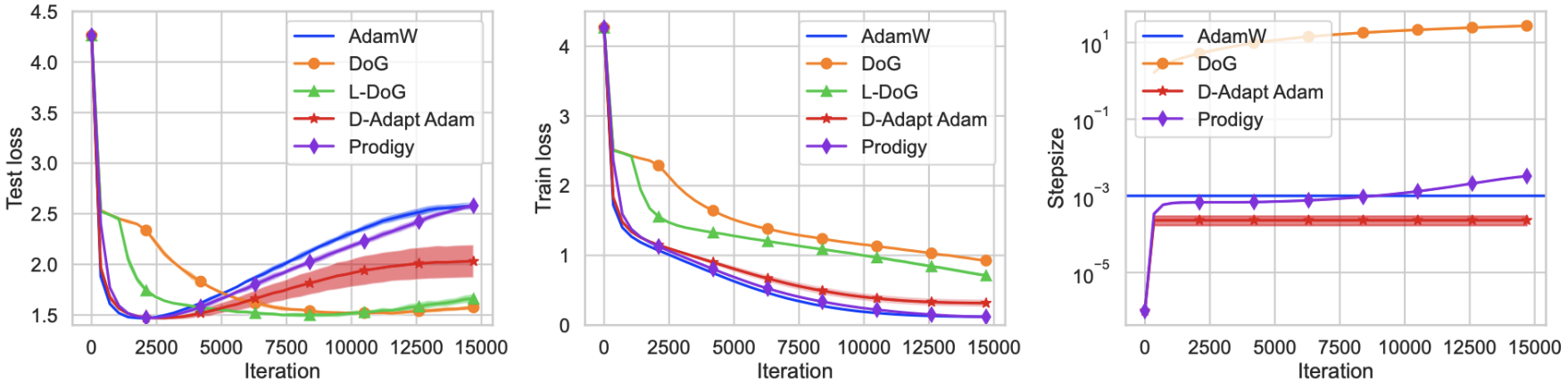


Figure 3: The test (left) and train (middle) loss curves as well as the estimated stepsize (right) when training a 6-layer nanoGPT transformer on the Shakespeare dataset.

Практические приложения

Так ли хороши parameter-free методы на практике? Помимо шага необходимо тюнить:

- weight decay
- momentum
- warm-up
- scheduler

В работе [13] провели бэнчмаркинг parameter-free оптимизаторов на AlgoPerf [7].

Практические приложения

Table 4: Best validation metrics over the course of training for baselines and naive learning-rate-free methods. Numbers colored in light gray indicate that the target was not achieved, while number colored in black indicate that the target was achieved. Validation metrics for naive learning-rate-free methods fail to beat NadamW baselines on any workload (top row). Up arrow indicates metric is maximized, down arrow minimized.

	Criteo 1TB	**fastMRI**	**ImageNet**		**LibriSpeech**		**OGBG**	**WMT**
	DLRMsmall	**U-Net**	**ResNet-50**	**ViT**	**Conformer**	**DeepSpeech**	**GNN**	**Transformer**
Metric	CE ↓	SSIM ↑	Error Rate ↓	Error Rate ↓	WER ↓	WER ↓	mAP ↑	BLEU ↑
AdamW	0.1235	0.722	0.252	0.222	0.085	0.118	**0.254**	29.64
NadamW	**0.1235**	**0.723**	**0.226**	**0.214**	**0.079**	**0.109**	0.253	**30.07**
COCOB	0.1238	0.719	0.305	0.274	0.172	0.149	0.234	26.67
D-Adapt (w. Adam)	0.1236	0.722	0.314	0.251	0.095	0.135	0.221	28.63
DoG	0.1334	0.714	0.317	0.275	0.467	0.226	0.231	26.34
DoWG	0.1259	0.715	0.389	0.998	0.844	0.839	0.205	0.00
Mechanic (w. Adam)	0.1240	0.722	0.321	0.249	0.098	0.124	0.235	1.82
Mechanic (w. Nadam)	0.1241	0.721	0.318	0.248	0.099	0.123	0.239	4.50
Mechanic	0.1361	0.714	0.301	0.263	0.183	0.209	0.045	11.94
MoMo	0.1254	0.716	0.313	0.268	0.116	0.154	0.236	25.78
MoMo (w. Adam)	0.1240	0.723	0.313	0.999	0.368	0.454	0.221	0.99
Prodigy	0.1241	0.723	0.317	0.251	0.098	0.149	0.212	0.57

Практические приложения

Table 5: Best validation metric values for AlgoPerf-calibrated methods achieved over the course of training. Numbers colored in light gray indicate that the target was not achieved, while number colored in black indicate that the target was achieved. Validation metrics for learning-rate-free methods with calibrated regularization and "decay schedule" are competitive with AdamW and NadamW baselines. Validation scores are much closer to baseline for all algorithms, and even beat the baseline on certain workloads.

	Criteo 1TB	**fastMRI**	**ImageNet**		**LibriSpeech**		**OGBG**	**WMT**
	DLRMsmall	**U-Net**	**ResNet-50**	**ViT**	**Conformer**	**DeepSpeech**	**GNN**	**Transformer**
Metric	CE ↓	SSIM ↑	Error Rate ↓	Error Rate ↓	WER ↓	WER ↓	mAP ↑	BLEU ↑
AdamW 33%	0.1235	0.722	0.252	0.222	0.085	0.118	0.254	29.64
AdamW 50%	0.1235	0.724	0.277	0.531	0.100	0.118	0.187	24.01
AdamW 66%	0.1234	0.722	0.234	0.236	0.075	0.108	0.257	29.60
NadamW 33%	0.1236	0.723	0.252	0.217	0.106	0.116	0.250	29.64
NadamW 50%	0.1234	0.722	0.251	0.232	0.085	0.115	0.276	30.85
NadamW 66%	0.1235	0.723	0.226	0.214	0.079	0.109	0.253	30.07
Mechanic (w. Adam) 33%	0.1240	0.724	0.238	0.225	0.084	0.113	0.245	29.91
Mechanic (w. Adam) 50%	0.1234	0.723	0.245	0.213	0.085	0.112	0.239	2.59
Mechanic (w. Adam) 66%	0.1235	0.723	0.275	0.222	0.082	0.111	0.245	25.53
Prodigy (w. Adam) 33%	0.1239	0.723	0.239	0.336	0.082	0.117	0.198	30.11
Prodigy (w. Adam) 50%	0.1234	0.722	0.223	0.213	0.087	0.107	0.244	29.94
Prodigy (w. Adam) 66%	0.1236	0.721	0.223	0.208	0.085	0.103	0.220	28.81

Практические приложения

Итого:

- Parameter-free методы учитывают только адаптивность размера шага.
- В реальных задачах нуждаются в необходимости тюнинга.
- Не имеют memory-efficient версий.

Sign-SGD

Алгоритм 3 SIGN-SGD [3]

Input: Start point $x^0 \in \mathbb{R}^d$, number of iterations T
Parameter: Stepsize $\gamma > 0$
for $t = 0, \ldots, T-1$ **do**
    $x^{t+1} = x^t - \gamma \text{sign}(\nabla f(x^t))$
end for

Алгоритм 4 Distributed SIGN-SGD [3]

Input: Start point $x^0 \in \mathbb{R}^d$, number of iterations T
Parameter: Stepsize $\gamma > 0$
for $t = 0, \ldots, T-1$ **do**
    $x^{t+1} = x^t - \gamma \text{sign}\left(\sum_{m=1}^{M} \text{sign}\left(\nabla f_m(x^t)\right)\right)$
end for

Connection to ADAM optimizer [1]

$$\widetilde{m}_t = \beta_1 \widetilde{m}_{t-1} + (1-\beta_1) g_t, \qquad m_t = \frac{\widetilde{m}_t}{1-\beta_1^{t+1}},$$

$$\widetilde{v}_t = \beta_2 \widetilde{v}_{t-1} + (1-\beta_2) g_t^2, \qquad v_t = \frac{\widetilde{v}_t}{1-\beta_2^{t+1}}.$$

Parameters update: $x^{t+1} = x^t - \alpha \frac{m_t}{\sqrt{v_t}}$.

Update direction representation:

$$\frac{m_t}{\sqrt{v_t}} = \frac{\operatorname{sign}(m_t)}{\sqrt{\frac{v_t}{m_t^2}}} = \sqrt{\frac{1}{1+\frac{v_t - m_t^2}{m_t^2}}} \odot \operatorname{sign}(m_t).$$

Convergence specialty

Assumption

The function f is L_∞-smooth, i.e., it satisfies $\|\nabla f(x) - \nabla f(y)\|_1 \leqslant L_\infty \|x - y\|_\infty$ *for any* $x, y \in \mathbb{R}^d$.

Sketch of the proof in deterministic case.

$$
\begin{aligned}
f(x^{t+1}) &\leqslant f(x^t) + \gamma \left\langle \nabla f(x^t), x^{t+1} - x^t \right\rangle + \frac{L_\infty}{2} \left\| x^{t+1} - x^t \right\|_\infty^2 \\
&\leqslant f(x^t) - \gamma \left\langle \nabla f(x^t), \operatorname{sign}\left(\nabla f(x^t)\right) \right\rangle \\
&\quad + \frac{L_\infty \gamma^2}{2} \left\| \operatorname{sign}\left(\nabla f(x^t)\right) \right\|_\infty^2 \\
&\leqslant f(x^t) - \gamma \left\| \nabla f(x^t) \right\|_1 + \frac{L_\infty \gamma^2}{2}.
\end{aligned}
$$

$\nabla f(x^t), x^{t+1}$

□

Convergence specialty

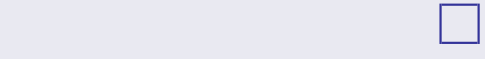
Continuing proof.

$$\frac{1}{T}\sum_{t=1}^{T-1}\left\|\nabla f(x^t)\right\|_1 \leqslant \frac{f(x^0)-f(x^*)}{\gamma T}+\frac{L_\infty \gamma}{2}.$$

Choosing $\gamma = \dfrac{\sqrt{f(x^0)-f(x^*)}}{\sqrt{L_\infty T}}$,

$$\frac{1}{T}\sum_{t=1}^{T-1}\left\|\nabla f(x^t)\right\|_1 = \mathcal{O}\left(\frac{\sqrt{\Delta^* L_\infty}}{\sqrt{T}}\right),$$

$$\left[\frac{1}{T}\sum_{t=1}^{T-1}\left\|\nabla f(x^t)\right\|_1\right]^2 = \mathcal{O}\left(\frac{\Delta^* L_\infty}{T}\right).$$



Comparison with SGD convergence

GD convergence: $\frac{1}{T}\sum_{t=1}^{T-1}\left\|\nabla f(x^t)\right\|_2^2 = \mathcal{O}\left(\frac{\Delta^* L_2}{T}\right)$;

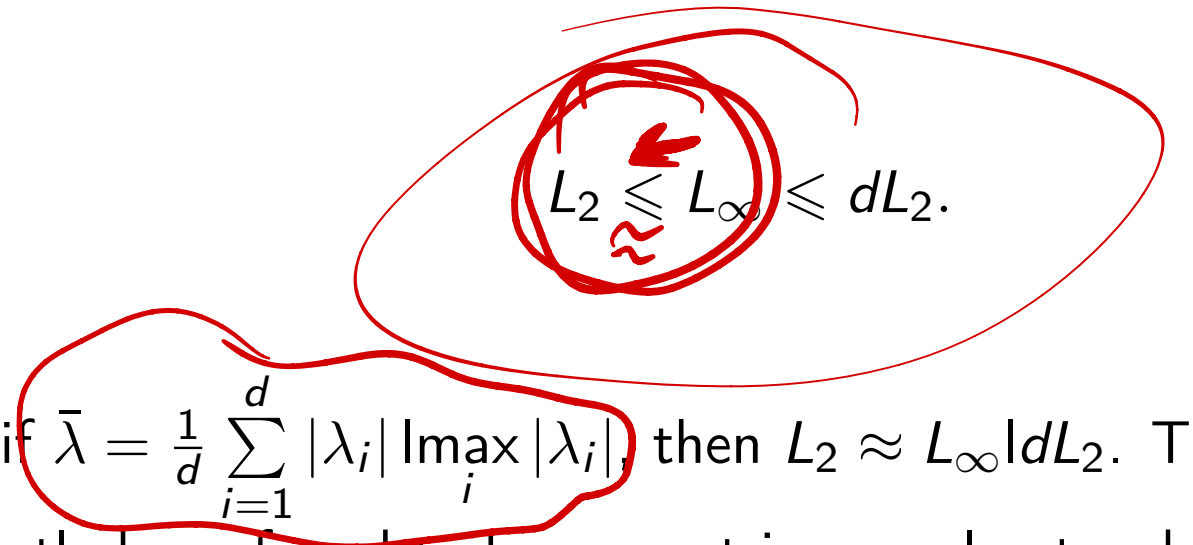
Sign-SGD convergence: $\left[\frac{1}{T}\sum_{t=1}^{T-1}\left\|\nabla f(x^t)\right\|_1\right]^2 = \mathcal{O}\left(\frac{\Delta^* L_\infty}{T}\right)$.

$\approx L_2$

$\left[\frac{1}{T}\sum_{t=0}^{T-1}\|\nabla f(x^t)\|_2\right]^2$

$\frac{1}{T}\sum_{t=0}^{T-1}\|\nabla f(x^t)\|_2^2$

Connection between L_∞ and L_2 [2]



However, if $\bar{\lambda} = \frac{1}{d}\sum_{i=1}^{d} |\lambda_i| \, \mathrm{l}\max_i |\lambda_i|$, then $L_2 \approx L_\infty \mathrm{l} dL_2$. This property have independently been found to be present in neural network training [6, 16, 10, 14].

Sign-SGD shortcomings [12]

- Sign-SGD does not converge even for linear functions with fixed stepsize.

Example

$$\min_{x\in[1,1]} \left[f(x) = \frac{1}{4}x = \frac{1}{4}(4x - x - x - x) \right].$$

$$g = \begin{cases} 4, & \text{with probability } \frac{1}{4} \\ -1, & \text{with probability } \frac{3}{4} \end{cases}.$$

$$\mathbb{E}_t\left[f(x^{t+1})\right] = \frac{1}{4}\left(x^t - \gamma\mathbb{E}\left[\text{sign}\,(g)\right]\right) = f(x^t) + \frac{\gamma}{8}.$$

Sign-SGD shortcomings [12]

- Sign-SGD does not converge for non-smooth and smooth convex functions even with adaptive stepsize.
- Sign-SGD with batch size 1 and stochastic gradients does not converge.

This forces us to use the gradient norm in the criterion even when analyzing convex objectives.

Parameter-free objectives

SGD requires stepsize $\gamma = \dfrac{\left\|x^0 - x^*\right\|}{M\sqrt{T}}$

to obtain $\mathcal{O}\left(\dfrac{\left\|x^0 - x^*\right\| M}{\sqrt{T}}\right)$ convergence rate.

Sign-SGD requires stepsize $\gamma = \dfrac{\sqrt{f(x^0) - f(x^*)}}{\sqrt{L_\infty T}}$

to obtain $\mathcal{O}\left(\dfrac{\sqrt{(f(x^0) - f(x^*))\, L_\infty}}{\sqrt{T}}\right)$ convergence rate.

We should utilize approximations of $\Delta^* = f(x^0) - f(x^*)$ and L_∞ in the stepsize γ.

ALIAS Sign-SGD [9]

Adaptation to $f(x^0) - f(x^*)$:

$$\widetilde{d}^t = \sum_{i=0}^{t-1} \gamma^i \langle \nabla f(x^{i+1}), \text{sign}(\nabla f(x^i)) \rangle.$$

Proof of $\Delta^* \geqslant \widetilde{d}^t$.

Starting from the convexity of the objective,

$$\begin{aligned} f(x^{t+1}) - f(x^t) &\leqslant \langle \nabla f(x^{t+1}), x^{t+1} - x^t \rangle \\ &= -\gamma^t \left\langle \nabla f(x^{t+1}), \text{sign}\left(\nabla f(x^t)\right) \right\rangle. \end{aligned}$$

Now we summarize both sides over the first n iterations:

$$\begin{aligned} -\Delta^* = f(x^*) - f(x^0) &\leqslant f(x^n) - f(x^0) = \sum_{t=0}^{n-1} f(x^{t+1}) - f(x^t) \\ &\leqslant -\sum_{t=0}^{n-1} \gamma^t \left\langle \nabla f(x^{t+1}), \text{sign}\left(\nabla f(x^t)\right) \right\rangle. \end{aligned}$$

ALIAS Sign-SGD

Continuing of the proof.

$$\widetilde{d}^n = \sum_{t=0}^{n-1} \gamma^t \left\langle \nabla f(x^{t+1}), \operatorname{sign}\left(\nabla f(x^t)\right) \right\rangle \leqslant \Delta^*.$$

□

There are second option, when

$$\widetilde{d}^t \equiv f(x^0) - \widetilde{f},$$

where $\widetilde{f} \leqslant f(x^*)$. $\widetilde{f} = 0$ is a valid estimate in empirical risk minimization settings. Moreover, $f(x^*) = 0$ for finding a point in the intersection of convex sets, completing positive semi-definite matrices, or solving convex inequalities [4].

ALIAS Sign-SGD

L_∞-approximation:

$$\lambda^t = \frac{1}{\sqrt{\sum_{i=0}^{t-1} \frac{\|\nabla f(x^{i+1}) - \nabla f(x^i)\|_1}{\|x^{i+1} - x^i\|_\infty}}}.$$

ALIAS Sign-SGD

Algorithm 2 ALIAS

Input: Starting point $x^0 \in \mathbb{R}^d$, initial L_∞-approximation $\eta^{-1} = 0$, initial Δ^*-approximation $d^0 \in \mathbb{R}_+$, lower bound $\widetilde{f}$ on $f(x^*)$, number of iterations T

for $t = 0, \ldots, T-1$ **do**

  Compute gradient $\nabla f(x^t)$

  $\eta^t = \eta^{t-1} + \frac{\|\nabla f(x^t) - \nabla f(x^{t-1})\|_1}{\|x^t - x^{t-1}\|_\infty}$; $\lambda^t = \frac{1}{\sqrt{\eta^t}}$

  if $t \neq 0$ **then**

    $\widetilde{d}^t = \sum_{i=0}^{t-1} \gamma^i \langle \nabla f(x^{i+1}), \operatorname{sign}(\nabla f(x^i)) \rangle$

    $d^t = \max\left(d^{t-1}, \widetilde{d}^t\right)$

  end if

  Option I: $\gamma^t = \lambda^t \sqrt{d^t}$

  Option II: $\gamma^t = \lambda^t \sqrt{f(x^0) - \widetilde{f}}$

  $x^{t+1} = x^t - \gamma^t \operatorname{sign}(\nabla f(x^t))$

end for

ALIAS Sign-SGD

Theorem

Suppose Assumptions of convexity and smoothness hold. We denote $\varepsilon = \frac{1}{T}\sum_{t=0}^{T-1} \|\nabla f(x^t)\|_1$. *Then ALIAS Sign-SGD Algorithm with Option I,* $d^0 < \Delta^*$ *to reach* ε*-accuracy needs*

$$\widetilde{\mathcal{O}}\left(\frac{(\Delta^*)^2 L_\infty}{d^0 \varepsilon^2}\right) \text{ iterations.}$$

Algorithm with Option II to reach ε*-accuracy needs*

$$\widetilde{\mathcal{O}}\left(\frac{\Delta^* L_\infty}{\varepsilon^2}\right) \text{ iterations.}$$

Proof. Starting from the convexity of the objective,

$$f(x^{t+1}) - f(x^t) \leqslant \langle \nabla f(x^{t+1}), x^{t+1} - x^t \rangle = -\gamma^t \left\langle \nabla f(x^{t+1}), \operatorname{sign}\left(\nabla f(x^t)\right) \right\rangle. \tag{43}$$

Now we summarize both sides over the first n iterations:

$$\begin{aligned} -\Delta^* = f(x^*) - f(x^0) &\overset{(i)}{\leqslant} f(x^n) - f(x^0) = \sum_{t=0}^{n-1} f(x^{t+1}) - f(x^t) \\ &\overset{43}{\leqslant} -\sum_{t=0}^{n-1} \gamma^t \left\langle \nabla f(x^{t+1}), \operatorname{sign}\left(\nabla f(x^t)\right) \right\rangle, \end{aligned}$$

where (i) is correct due to Assumption 3.3. Changing the sign of the inequality,

$$\widetilde{d}^n = \sum_{t=0}^{n-1} \gamma^t \left\langle \nabla f(x^{t+1}), \operatorname{sign}\left(\nabla f(x^t)\right) \right\rangle \leqslant \Delta^*.$$

Since our algorithm performs $d^n = \max\left(d^{n-1}, \widetilde{d}^n\right)$ and we initialize our sequence with $d^0 < \Delta^*$, we obtain the required statement. □

Lemma F.2 (Descent lemma)**.** *For Algorithm 2 under Assumptions 3.1, 3.2, 3.3, 3.4, the following estimate is valid:*

$$\sum_{t=0}^{T-1} \gamma^t \left\|\nabla f(x^t)\right\|_1 \leqslant \Delta^* + \sum_{t=0}^{T-1} (\gamma^t)^2 L_\infty^t,$$

where $L_\infty^t = \frac{\left\|\nabla f(x^{t+1}) - \nabla f(x^t)\right\|_1}{\left\|x^{t+1} - x^t\right\|_\infty}$.

Proof.

$$\begin{aligned} f(x^{t+1}) &\leqslant f(x^t) + \left\langle \nabla f(x^{t+1}), x^{t+1} - x^t \right\rangle = f(x^t) - \gamma^t \left\langle \nabla f(x^{t+1}), \operatorname{sign}\left(\nabla f(x^t)\right) \right\rangle \\ &= f(x^t) - \gamma^t \left\|\nabla f(x^t)\right\|_1 - \gamma^t \left\langle \nabla f(x^{t+1}) - \nabla f(x^t), \operatorname{sign}\left(\nabla f(x^t)\right) \right\rangle \\ &\overset{Conj}{\leqslant} f(x^t) - \gamma^t \left\|\nabla f(x^t)\right\|_1 + \gamma^t \left\|\nabla f(x^{t+1}) - \nabla f(x^t)\right\|_1 \left\|\operatorname{sign}\left(\nabla f(x^t)\right)\right\|_\infty \\ &\leqslant f(x^t) - \gamma^t \left\|\nabla f(x^t)\right\|_1 + \gamma^t \left\|\nabla f(x^{t+1}) - \nabla f(x^t)\right\|_1 \\ &\overset{(i)}{=} f(x^t) - \gamma^t \left\|\nabla f(x^t)\right\|_1 + \gamma^t \frac{\left\|\nabla f(x^{t+1}) - \nabla f(x^t)\right\|_1}{\left\|x^{t+1} - x^t\right\|_\infty} \left\|x^{t+1} - x^t\right\|_\infty \\ &= f(x^t) - \gamma^t \left\|\nabla f(x^t)\right\|_1 + (\gamma^t)^2 \frac{\left\|\nabla f(x^{t+1}) - \nabla f(x^t)\right\|_1}{\left\|x^{t+1} - x^t\right\|_\infty}, \end{aligned}$$

where in (i) we assume $\left\|x^{t+1} - x^t\right\|_\infty \neq 0$. Indeed, $\left\|x^{t+1} - x^t\right\|_\infty = 0$ follows from the equality $\operatorname{sign}\left(\nabla f(x^t)\right) = 0$, which means that we find the optimum and do need to find another point x^{t+1}. Now we denote $L_\infty^t = \frac{\left\|\nabla f(x^{t+1}) - \nabla f(x^t)\right\|_1}{\left\|x^{t+1} - x^t\right\|_\infty}$. Summing over all iterations, we obtain

$$\begin{aligned} \sum_{t=0}^{T-1} \gamma^t \left\|f(x^t)\right\|_1 &\leqslant \sum_{t=0}^{T-1} \left[f(x^t) - f(x^{t+1})\right] + \sum_{t=0}^{T-1} (\gamma^t)^2 L_\infty^t \\ &= f(x^0) - f(x^*) + \sum_{t=0}^{T-1} (\gamma^t)^2 L_\infty^t \leqslant \Delta^* + \sum_{t=0}^{T-1} (\gamma^t)^2 L_\infty^t, \end{aligned}$$

which ends the proof of the lemma. □

Theorem F.3 (Theorem 3.5). *Suppose Assumptions 3.1, 3.2, 3.3, 3.4 hold. We denote* $\varepsilon \geqslant \frac{1}{T}\sum_{t=0}^{T-1} \|\nabla f(x^t)\|_1, L_\infty^t = \frac{\left\|\nabla f(x^{t+1}) - \nabla f(x^t)\right\|_1}{\left\|x^{t+1} - x^t\right\|_\infty}$. *Then Algorithm 2 with Option I,* $d^0 < \Delta^*$ *to reach* ε*-accuracy needs*

$$\widetilde{\mathcal{O}}\left(\frac{(\Delta^*)^2 (L_\infty)^3}{d^0 (L_\infty^0)^2 \varepsilon^2}\right) \quad \textit{iterations.}$$

Algorithm 2 with Option II to reach ε-accuracy needs

$$\widetilde{\mathcal{O}}\left(\frac{\Delta^* \left(L_\infty\right)^3}{\left(L_\infty^0\right)^2 \varepsilon^2}\right) \text{ iterations.}$$

Proof. Let us start with the result of Lemma F.2:

$$\sum_{t=0}^{T-1} \gamma^t \|\nabla f(x^t)\|_1 \leqslant \Delta^* + \sum_{t=0}^{T-1} (\gamma^t)^2 L_\infty^t. \tag{44}$$

Now we use our γ^t choice. Let us firstly estimate the denominator that is exactly $\lambda^t = \frac{1}{\sqrt{\sum\limits_{i=0}^{t-1} \frac{\|\nabla f(x^{i+1}) - \nabla f(x^i)\|_1}{\|x^{i+1} - x^i\|_\infty}}} = \frac{1}{\sqrt{\sum\limits_{i=0}^{t-1} L_\infty^i}}$ and is the same for both Options I and II. Let us estimate the following term.

$$\sum_{t=0}^{T-1} (\lambda^t)^2 L_\infty^t = \sum_{t=0}^{T-1} \frac{L_\infty^t}{\sum\limits_{i=0}^{t-1} L_\infty^i}.$$

We mention, that each L_∞^i is bounded from the definition of smoothness (see Assumption 3.1), i.e., $L_\infty^i \leqslant L_\infty$. We consider the sequence $\left\{L_\infty^i\right\}_{i=0}^{T-1}$. Since each term in this sequence is bounded, there exists r such that $\sum\limits_{i=0}^{r-2} L_\infty^i \leqslant L_\infty^{r-1}$ and for each $t \geqslant r-1$ such that $\sum\limits_{i=0}^{t} L_\infty^i \geqslant L_\infty^{t+1}$. In that way, we divide the sum into two parts:

$$\sum_{t=0}^{T-1} \frac{L_\infty^t}{\sum\limits_{i=0}^{t-1} L_\infty^i} = \sum_{t=0}^{r-1} \frac{L_\infty^t}{\sum\limits_{i=0}^{t-1} L_\infty^i} + \sum_{t=r}^{T-1} \frac{L_\infty^t}{\sum\limits_{i=0}^{t-1} L_\infty^i}. \tag{45}$$

Considering the first sum in equation 45, we mention, that we can estimate the denominator as $\sum_{i=0}^{t-1} L_\infty^i \geqslant L_\infty^0$. As for the numerator. Thus,

$$\sum_{t=0}^{r-1} \frac{L_\infty^t}{\sum\limits_{i=0}^{t-1} L_\infty^i} \leqslant \frac{1}{L_\infty^0} \left(\sum_{t=0}^{r-2} L_\infty^t + L_\infty^{r-1}\right) \leqslant \frac{2L_\infty^{r-1}}{L_\infty^0} \leqslant \frac{2L_\infty}{L_\infty^0}. \tag{46}$$

Considering the second sum in equation 45, we have

$$\sum_{t=r}^{T-1} \frac{L_\infty^t}{\sum\limits_{i=0}^{t-1} L_\infty^i} = \sum_{t=r}^{T-1} \frac{L_\infty^t}{\frac{1}{2}\sum\limits_{i=0}^{t-1} L_\infty^i + \frac{1}{2}\sum\limits_{i=0}^{t-1} L_\infty^i}.$$

Estimating any of the sums in the denominator, we claim, that $\sum\limits_{i=0}^{t-1} L_\infty^i \geqslant L_\infty^t$, since $t-1 \geqslant r-1$. In that way,

$$\sum_{t=r}^{T-1} \frac{L_\infty^t}{\sum\limits_{i=0}^{t-1} L_\infty^i} \leqslant \sum_{t=r}^{T-1} \frac{2L_\infty^t}{\sum\limits_{i=0}^{t} L_\infty^i} \leqslant 2\sum_{t=0}^{T-1} \frac{L_\infty^t}{\sum\limits_{i=0}^{t} L_\infty^i}. \tag{47}$$

Next we denote $s^t = \sum\limits_{i=0}^{t} L_\infty^t$ and have

$$L_\infty^t \frac{1}{\sum\limits_{i=0}^{t} L_\infty^i} = (s^t - s^{t-1}) \frac{1}{\sum\limits_{i=0}^{t} L_\infty^i} = \int\limits_{s^{t-1}}^{s^t} \frac{1}{\sum\limits_{i=0}^{t} L_\infty^i} dx \overset{(i)}{\leqslant} \int\limits_{s^{t-1}}^{s^t} \frac{1}{x} dx, \tag{48}$$

where (i) was done due to $\frac{1}{x}$ is a non-increasing function on $(0, +\infty)$. Summing over t, we obtain

$$2 \sum_{t=1}^{T} \frac{L_\infty^t}{\sum\limits_{i=0}^{t} L_\infty^i} \leqslant 2 \int\limits_{s^0}^{s^T} \frac{1}{x} dx = 2 \log(s^T) - 2 \log(s^0) = 2 \log \left(\frac{\sum\limits_{t=1}^{T} L_\infty^t}{L_\infty^0} \right) \leqslant 2 \log \left(\frac{L_\infty T}{L_\infty^0} \right).$$

Combining this estimate with equation 47,

$$\sum_{t=r}^{T-1} \frac{L_\infty^t}{\sum\limits_{i=0}^{t-1} L_\infty^i} \leqslant 2 \sum_{t=1}^{T} \frac{L_\infty^t}{\sum\limits_{i=0}^{t} L_\infty^i} + 2 \leqslant 2 \left(\log \left(\frac{L_\infty T}{L_\infty^0} \right) + 1 \right) \leqslant 4 \log \left(\frac{L_\infty T}{L_\infty^0} \right). \tag{49}$$

Substituting equation 46 and equation 49 into equation 45, we obtain

$$\sum_{t=0}^{T-1} (\lambda^t)^2 L_\infty^t \leqslant 2 \frac{L_\infty}{L_\infty^0} + 4 \log \left(\frac{L_\infty T}{L_\infty^0} \right). \tag{50}$$

We additionally note, that if $r > T - 1$, only first term remains in this estimate, consequently our bound equation 50 is correct.

In this way, utilizing Option I from Algorithm 2, equation 44 together with equation 50 yields

$$\begin{aligned}
\sqrt{d^0} \lambda^{T-1} \sum_{t=0}^{T-1} \|\nabla f(x^t)\|_1 &\overset{(i)}{\leqslant} \sum_{t=0}^{T-1} \sqrt{d^t} \lambda^t \|\nabla f(x^t)\|_1 \leqslant \Delta^* + \sum_{t=0}^{T-1} d^t (\lambda^t)^2 L_\infty^t \\
&\overset{\text{Lemma F.1}}{\leqslant} \Delta^* + \Delta^* \sum_{t=0}^{T-1} (\lambda^t)^2 L_\infty^t, \\
\sum_{t=0}^{T-1} \|\nabla f(x^t)\|_1 &\leqslant \frac{\Delta^*}{\sqrt{d^0} \lambda^{T-1}} + \frac{\Delta^*}{\sqrt{d^0} \lambda^{T-1}} \sum_{t=0}^{T-1} (\lambda^t)^2 L_\infty^t \\
&\overset{50}{\leqslant} \frac{\Delta^*}{\sqrt{d^0} \lambda^{T-1}} + 4 \frac{\Delta^*}{\sqrt{d^0} \lambda^{T-1}} \log \left(\frac{L_\infty T}{L_\infty^0} \right) + 2 \frac{\Delta^* L_\infty}{\sqrt{d^0} \lambda^{T-1} L_\infty^0} \\
&\leqslant 7 \frac{\Delta^* L_\infty}{\sqrt{d^0} \lambda^{T-1} L_\infty^0} \log \left(\frac{L_\infty T}{L_\infty^0} \right),
\end{aligned} \tag{51}$$

where (i) was done due to the fact that d^0 is minimal from all $\{d^t\}_{t=0}^{T-1}$ (Line 7 from Algorithm 2) and the definition of λ^t. Utilizing $\frac{1}{\lambda^{T-1}} = \sqrt{\sum\limits_{t=0}^{T-2} L_\infty^t} \leqslant \sqrt{L_\infty T}$, we obtain the final estimate

$$\frac{1}{T} \sum_{t=0}^{T-1} \|\nabla f(x^t)\|_1 \leqslant \frac{7 \Delta^* (L_\infty)^{\frac{3}{2}}}{\sqrt{d^0 T} L_\infty^0} \log \left(\frac{L_\infty T}{L_\infty^0} \right).$$

Expressing the number of iterations and using $\varepsilon \geqslant \frac{1}{T} \sum\limits_{t=0}^{T-1} \|\nabla f(x^t)\|_1$ as a criterion, we obtain that the algorithm needs $\widetilde{\mathcal{O}} \left(\frac{(\Delta^*)^2 (L_\infty)^3}{d^0 (L_\infty^0)^2 \varepsilon^2} \right)$ iterations to reach ε-accuracy.

Considering Option II from Algorithm 2, we can proceed absolutely analogical, however, using $f(x^0) - \widetilde{f} \geqslant \Delta^*$ instead of Lemma F.1. In that way,

$$\begin{aligned}\frac{1}{T}\sum_{t=0}^{T-1}\|\nabla f(x^t)\|_1 &\leqslant \frac{\Delta^*\sqrt{L_\infty}}{\sqrt{(f(x^0)-\widetilde{f})T}} + \frac{4(f(x^0)-\widetilde{f})\sqrt{L_\infty}}{\sqrt{(f(x^0)-\widetilde{f})T}}\log\left(\frac{L_\infty T}{L_\infty^0}\right) \\ &\quad + \frac{2(f(x^0)-\widetilde{f})\,(L_\infty)^{\frac{3}{2}}}{\sqrt{(f(x^0)-\widetilde{f})T L_\infty^0}} \\ &\leqslant \frac{7\sqrt{(f(x^0)-\widetilde{f})}\,(L_\infty)^{\frac{3}{2}}}{\sqrt{T}L_\infty^0}\log\left(\frac{L_\infty T}{L_\infty^0}\right).\end{aligned}$$

Expressing the number of iterations, using $\varepsilon \geqslant \frac{1}{T}\sum\limits_{t=0}^{T-1}\|\nabla f(x^t)\|_1$ as a criterion, and utilizing $\widetilde{f}$ is an approximation of $f(x^*)$, we obtain that the algorithm needs $\widetilde{\mathcal{O}}\left(\frac{\Delta^*(L_\infty)^3}{(L_\infty^0)^2\varepsilon^2}\right)$ iterations to reach ε-accuracy. □

Remark F.4 (Remark 3.6). Under conditions of Theorem 3.5 Algorithm 2 with $\lambda^t = \frac{1}{\sqrt{L_\infty + \sum\limits_{i=0}^{t-1}\frac{\|\nabla f(x^{i+1})-\nabla f(x^i)\|_1}{\|x^{i+1}-x^i\|_\infty}}}$ and Option II to reach ε-accuracy needs

$$\widetilde{\mathcal{O}}\left(\frac{\Delta^* L_\infty}{\varepsilon^2}\right) \text{ iterations,}$$

where $\varepsilon \geqslant \frac{1}{T}\sum\limits_{t=0}^{T-1}\|\nabla f(x^t)\|_1$.

Proof. The proof of the remark repeats the proof of Theorem 3.5 except for the estimate on $\sum\limits_{t=0}^{T-1}(\lambda^t)^2 L_\infty^t$ term. Let us derive it. We use definition $L_\infty^t = \frac{\|\nabla f(x^{t+1})-\nabla f(x^t)\|_1}{\|x^{t+1}-x^t\|_\infty}$.

$$\sum_{t=0}^{T-1}(\lambda^t)^2 L_\infty^t = \sum_{t=0}^{T-1}\frac{L_\infty^t}{L_\infty + \sum\limits_{i=0}^{t-1}L_\infty^i} \leqslant \sum_{t=0}^{T-1}\frac{L_\infty^t}{\sum\limits_{i=0}^{t}L_\infty^i}.$$

Continuing analogically to equation 48 - equation 49, we get

$$\sum_{t=0}^{T-1}(\lambda^t)^2 L_\infty^t \leqslant 2\log\left(\frac{L_\infty T}{L_\infty^0}\right).$$

Substituting this bound into equation 51 instead of equation 50, we ends the proof of the remark. □

F.2 Stochastic gradient setting

In this section we denote $g_{\xi^t}^t$ the stochastic gradient at the t-th iteration (point x^t), according to the stochastic realization ξ^t at the t-th iteration.

Before proceeding to the theoretical analysis of the algorithm, we present its formal description, Algorithm 7, specifying which option for the sequence d^t we use in practice and which one we analyze theoretically.

Adam version

Algorithm 3 ALIAS Adam version

Input: Start points $x^{-1}, x^0 \in \mathbb{R}^d$, $r^0, m^0, v^0 = 0$, $d^{-1} > 0$, number of iterations T

Parameters: $\gamma^t, \beta_1, \beta_2 > 0$

for $t = 0, \ldots, T-1$ **do**

$$r^{t+1} = \sqrt{\beta_2} r^t + \left(1 - \sqrt{\beta_2}\right) d^{t-1} \left\langle g^t_{\xi^t}, \operatorname{sign}(g^{t-1}_{\xi^{t-1}}) \right\rangle$$

$$d^t = \max\left\{d^{t-1}, r^{t+1}\right\}$$

$$m^{t+1} = \beta_1 m^t + (1-\beta_1) d^t g^t_{\xi^t}$$

$$v^{t+1} = \beta_2 v^t + (1-\beta_2)\left(d^t\right)^2 \left(g^t_{\xi^t}\right)^2$$

$$x^{t+1} = x^t - \gamma^t \sqrt{\frac{(d^t)^2}{1 + \frac{v^{t+1} - (m^{t+1})^2}{(m^{t+1})^2}}} \odot \operatorname{sign}(m^{t+1})$$

end for

Experiments

Table 1: SIGN methods on LLAMA pre-training.

Algorithm	Validation Loss (↓)	Perplexity (↓)
SIGN-SGD (lr)	3.041	20.923
SIGN-SGD (lr, cosine sc)	2.992	19.923
STEEPEST DESCENT (lr, cosine sc)	3.035	20.791
NORMALIZED SGD (lr, cosine sc)	3.135	22.982
ALIAS (**ours**)	3.017	20.422
SIGN-SGD (wd, lr)	3.041	20.923
SIGN-SGD (wd, lr, cosine sc)	2.980	19.693
STEEPEST DESCENT (wd, lr, cosine sc)	3.022	20.537
NORMALIZED SGD (wd, lr, cosine sc)	3.006	20.169
ALIAS (wd) (**ours**)	3.006	20.169

Table 2: SIGN-SGD methods with added momentum parameter (β), ADAMW and PRODIGY on LLAMA pre-training.

Algorithm	Validation Loss (↓)	Perplexity (↓)
SIGN-SGD (wd, β, lr)	2.968	19.459
SIGN-SGD (wd, β, lr, cosine sc)	2.923	18.596
STEEPEST DESC. (wd, β, lr, cosine sc)	2.932	18.765
NORM. SGD (wd, β, lr, cosine sc)	2.934	18.803
ADAMW (wd, β, lr, cosine sc)	2.929	18.698
PRODIGY (wd, β)	3.003	20.145
PRODIGY (wd, β, cosine sc)	2.930	18.727
ALIAS Adam version (wd, β) (**ours**)	2.976	19.609
ALIAS Adam version (wd, β, cosine sc) (**ours**)	2.918	18.504

Experiments

Table 7: Parameter-free methods on LLAMA pre-training.

Algorithm	Validation Loss (↓)	Perplexity (↓)
DoG	2.939	18.897
D-ADAPTATION (with Adam)	2.927	18.672
MoMo (with Adam)	2.925	18.634
PRODIGY	2.930	18.727
ALIAS Adam version (wd) (**ours**)	2.918	18.504

Table 11: Parameter-free methods and ADAMW on MRI reconstruction and molecular property prediction tasks.

Algorithm	MRI, SSIM (↑)	MPP, mAP (↑)
ADAMW	0.723	0.254
DoG	0.714	0.231
D-ADAPTATION (with Adam)	0.722	0.221
MoMo (with Adam)	0.723	0.221
PRODIGY	0.723	0.212
ALIAS Adam version (wd) (**ours**)	0.724	0.242

Experiments

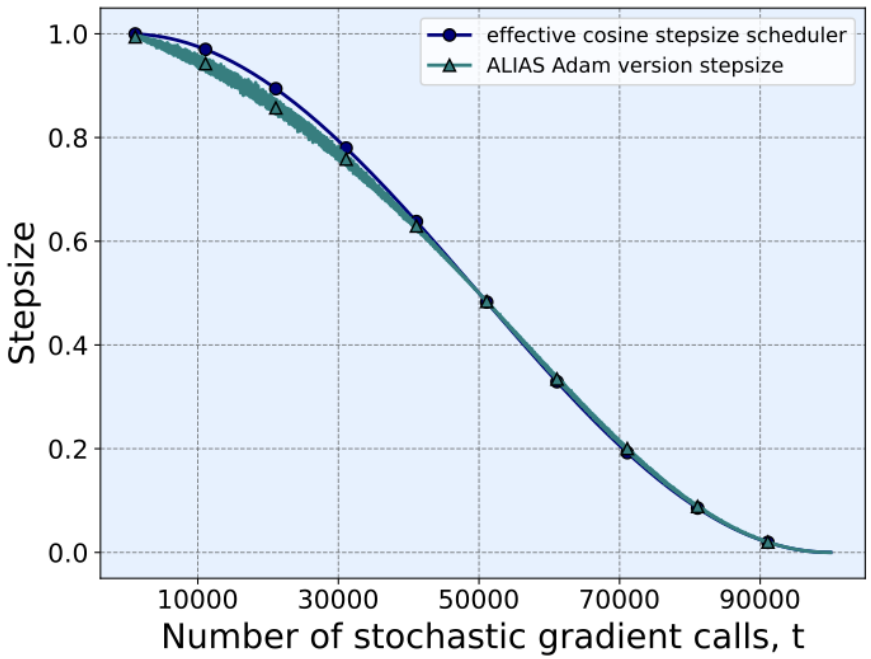


Figure 2: Comparison of ALIAS Adam version stepsize with constant γ^t with effective cosine stepsize scheduler.

Thank you for your attention!

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